

ON SOME INEQUALITIES SATISFIED BY BETA AND GAMMA FUNCTIONS

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SUMMARY

By using hypercomplex multivariate normal distribution theory and Bellman-Wishart distribution, Bellman (1956), an inequality for gamma functions is derived. This inequality generalizes a recent result of Selliah (1976). Similarly by using Olkin's (1959) modified multivariate beta distribution, an inequality for beta functions is derived and a result for Gauss' hypergeometric series of matrix argument is given.

1. Introduction

Selliah (1976) proves that

$$(1) \quad [\Gamma_p^2(\delta + \alpha)/\Gamma_p(\delta)\Gamma_p(\delta + 2\alpha)] \leq \delta/(\delta + p\alpha^2),$$

where $\delta + \alpha > 2t(p-1) + 1$, $\delta > 0$, $p > 0$, $t = \frac{1}{4}$, and

$$(2) \quad \Gamma_p(\delta) = \pi^{tp(p-1)} \prod_{i=1}^p \Gamma(\delta - 2t(i-1)).$$

although Selliah (1976) proves the result (1) for $t = \frac{1}{4}$, the result (1) is true for $t = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$. However, the purpose of this paper is not to prove that (1) is true for $t = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$, but to prove a far more general result from which (1) follows as a particular case. For this purpose we need the theory of hypercomplex normal multivariate distributions given by Kabe (1976). We state this theory as follows. Let $X_1, X_2, \dots, X_{4t}$, $t = \frac{1}{4}, \frac{1}{2}, 1, 2$ be $4t \times N$ real random matrices having a $4pNt$ variate normal density

$$(3) \quad (4\pi t)^{-2pNt} |\Sigma_0|^{-\frac{1}{2}N} \exp \{-\text{tr} \Sigma_0^{-1} (X - \mu)(X - \mu)'/4t\}$$

where

$$(4) \quad X = [X_1', \dots, X_{4t}']', \text{ and } \Sigma_0 \text{ is}$$

$$(5) \quad \begin{bmatrix} \Sigma_1 & -\Sigma_2 & -\Sigma_3 & -\Sigma_4 & -\Sigma_5 & -\Sigma_6 & -\Sigma_7 & -\Sigma_8 \\ \Sigma_2 & \Sigma_1 & -\Sigma_4 & \Sigma_3 & -\Sigma_6 & \Sigma_5 & \Sigma_8 & -\Sigma_7 \\ \Sigma_3 & \Sigma_4 & \Sigma_1 & -\Sigma_2 & -\Sigma_7 & \Sigma_8 & \Sigma_5 & \Sigma_6 \\ \Sigma_4 & -\Sigma_3 & \Sigma_2 & \Sigma_1 & -\Sigma_8 & \Sigma_7 & -\Sigma_6 & \Sigma_5 \\ \Sigma_5 & \Sigma_6 & \Sigma_7 & \Sigma_8 & \Sigma_1 & -\Sigma_2 & -\Sigma_3 & -\Sigma_4 \\ \Sigma_6 & -\Sigma_5 & \Sigma_8 & -\Sigma_7 & \Sigma_2 & \Sigma_1 & \Sigma_4 & \Sigma_3 \\ \Sigma_7 & -\Sigma_8 & -\Sigma_5 & \Sigma_6 & \Sigma_3 & -\Sigma_4 & \Sigma_1 & \Sigma_2 \\ \Sigma_8 & \Sigma_7 & -\Sigma_6 & -\Sigma_5 & \Sigma_4 & \Sigma_3 & -\Sigma_2 & \Sigma_1 \end{bmatrix}$$

further

$$(6) \quad \mu = \mu_0 [M'_1, M'_2, \dots, M'_{4t}], \text{ and } 4t \times 4t \mu_0 \text{ is}$$

$$(7) \quad \begin{bmatrix} \mu_1 & -\mu_2 & -\mu_3 & -\mu_4 & -\mu_5 & -\mu_6 & -\mu_7 & -\mu_8 \\ \mu_2 & \mu_1 & -\mu_4 & \mu_3 & -\mu_6 & \mu_5 & \mu_8 & \mu_7 \\ \mu_3 & \mu_4 & \mu_1 & -\mu_2 & -\mu_7 & \mu_8 & \mu_5 & \mu_6 \\ \mu_4 & -\mu_3 & \mu_2 & \mu_1 & -\mu_8 & \mu_7 & -\mu_6 & \mu_5 \\ \mu_5 & \mu_6 & \mu_7 & \mu_8 & \mu_1 & -\mu_2 & -\mu_3 & -\mu_4 \\ \mu_6 & -\mu_5 & \mu_8 & -\mu_7 & \mu_2 & \mu_1 & \mu_4 & \mu_3 \\ \mu_7 & -\mu_8 & -\mu_5 & \mu_6 & \mu_3 & -\mu_4 & \mu_1 & \mu_2 \\ \mu_8 & \mu_7 & -\mu_6 & -\mu_5 & \mu_4 & \mu_3 & -\mu_2 & \mu_1 \end{bmatrix}$$

The above pattern for Σ_0 (and μ_0) is for $t = 2$, i.e. for the octonion case. When $t = 1$ we have the quaternion case and in this case the pattern for Σ_0 (and μ_0) is given by the first four rows and first four columns of Σ_0 (and μ_0). When $t = \frac{1}{2}$, we have the complex case and the pattern for Σ_0 (and μ_0) is given by the first two rows and columns of Σ_0 (and μ_0). When $t = \frac{1}{4}$ we have the real case and the pattern for Σ_0 (and μ_0) is trivially given by the first row and column of Σ_0 (and μ_0). We consider the case $t = 2$. Here Σ_1 is a $p \times p$ real positive definite symmetric matrix; $\Sigma_2, \dots, \Sigma_8$ are real $p \times p$ skew symmetric matrices; μ_j $p \times q$, M_j $q \times N$, $j = 1, \dots, 8$ are given real matrices.

Now we set $Y = X_1 + iX_2 + jX_3 + kX_4 + lX_5 + mX_6 + nX_7 + oX_8$, where the special (generator) octonions i, j, k, l, m, n, o satisfy the multiplication rules

$$(8) \quad i^2 = j^2 = k^2 = l^2 = m^2 = n^2 = o^2 = -1 = ijk = iln = ion = jln = jmo = klo = knm$$

and we find $\bar{Y} = X_1 - iX_2 - jX_3 - kX_4 - lX_5 - mX_6 - nX_7 - oX_8$ as the octonion conjugate of Y .

After some matrix algebra similar to the complex case, Goodman (1963), may write down the pN variate octonion normal density of Y as

$$(9) \quad \pi^{-2pNt} |\Sigma|^{-2Nt} \exp \{-\text{tr} \Sigma^{-1} (Y - \mu M) (\overline{Y - \mu M})'\}, \text{ where}$$

$$(10) \quad \Sigma = \Sigma_1 + i\Sigma_2 + j\Sigma_3 + k\Sigma_4 + l\Sigma_5 + m\Sigma_6 + n\Sigma_7 + o\Sigma_8,$$

$$(11) \quad \mu = \mu_1 + i\mu_2 + j\mu_3 + k\mu_4 + l\mu_5 + m\mu_6 + n\mu_7 + o\mu_8,$$

$$(12) \quad M = M_1 + iM_2 + jM_3 + kM_4 + lM_5 + mM_6 + nM_7 + oM_8.$$

We note that $\bar{\Sigma}' = \Sigma$, i.e., Σ is an octonion positive definite Hermitian (opdH) matrix, and Σ has real roots.

The biocotnion case is just a complex copy of the real octonion case. Let $Z_1, Z_2, \dots, Z_8$ be eight $p \times N$ complex matrices, then the biocotnion matrix Z is defined by

$$(13) \quad Z = Z_1 + iZ_2 + jZ_3 + kZ_4 + lZ_5 + mZ_6 + nZ_7 + oZ_8,$$

where $Z_1, \dots, Z_8$ have a joint $8pN$ variate complex normal density similar to (3) with a joint covariance matrix similar to (5), where now Σ_1 is $p \times p$ positive definite Hermitian and $\Sigma_2, \dots, \Sigma_8$ are skew Hermitian $p \times p$ matrices; μ_0 has the same structure as (7) and μ has the same structure as (6). The pN variate bioctonion normal density of Z is

$$(14) \quad \pi^{-4pNt} |\Sigma|^{-4Nt} \exp \{-\text{tr} \Sigma^{-1} (Z - \mu M) (\overline{Z - \mu M})'\},$$

where the structure of Σ, μ, M in (14) is the same as that in (10), (11), (12), except that Σ_1 is Hermitian, etc. Note that in bioctonion case also $\bar{\Sigma}' = \Sigma$, i.e., Σ is a bioctonion positive definite Hermitian (BopdH) matrix and Σ has real roots.

Although sometimes the same symbol denotes different quantities, its meaning is made explicit in the context. When $t = 1$ in (14), (14) gives a biquaternion case and when $t = \frac{1}{2}$ in (14), (14) gives the complex case.

Now by assuming $\mu = 0$ in (9), Kabe (1976) finds the Wishart density, with N degrees of freedom (d.f.), of the opdH matrix $S = Y\bar{Y}'$ as

$$(15) \quad (\Gamma_p(\delta))^{-1} |\Sigma|^{-\delta} \exp \{-\text{tr} \Sigma^{-1} S\} |S|^{\delta-2t(p-1)-1},$$

where $\delta = 2Nt$. Again by assuming $\mu = 0$ in (14), Kabe (1976) obtains the Wishart den-

sity with N d.f. of the BopdH matrix $S = ZZ'$ as

$$(16) (\Gamma_p(\delta))^{-1} |\Sigma|^{-\delta} \exp \{-\text{tr} \Sigma^{-1} S\} |S|^{\delta - 4t(p-1) - 1},$$

where in (16) $\delta = 4Nt$ and

$$(17) \Gamma_p(\delta) = \pi^{2tp(p-1)} \prod_{i=1}^p \Gamma(\delta - 4t(i-1)).$$

The Bellman-Wishart density of a pxp opdH matrix S with N d.f. is defined as follows. Let S be a pxp opdH matrix and

$$(18) S_k = (S_{ij}), 1 \leq i, j \leq k; S^{(k)} = (S_{ij}), k \leq i, j \leq p,$$

and for convenience $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$. Then the Bellman-Wishart density of S is $\phi(S, \Lambda) = \phi$

$$(19) (\Gamma_p^*(a))^{-1} \prod_{i=1}^p \lambda_i^{-a_{p-i+1}} |S|^{\dot{a}_p - 2t(p-1) - 1} \exp \{-\text{tr} \Lambda^{-1} S\} \prod_{j=2}^p |S_k|^{-k_{i-1}},$$

where $a_p = 2Nt$, $a_j = k_{p-j+1} + \dots + k_p$, $k = (k_1, \dots, k_p)$, $a = (a_1, \dots, a_p)$ and $a_j > 2t(p-1) + 1$, and

$$(20) \Gamma_p^*(a) = \pi^{tp(p-1)} \prod_{i=1}^p \Gamma(a_i - 2t(i-1)).$$

Similarly, when S is a pxp BopdH matrix in (19) we replace t by $2t$ and the opdH S by the BopdH S and (19) yields the Bell-Wishart density with N d.f. of the pxp BopdH matrix S .

If a scalar variate T unbiasedly estimates $\psi(\lambda_1, \dots, \lambda_p) = \psi$, then the multiparameter Cramer-Rao bound is

$$(21) V(T) \geq \sum_{i=j}^p \sum_{j=1}^p J_{ij} \frac{\partial \psi}{\partial \lambda_i} \frac{\partial \psi}{\partial \lambda_j},$$

where

$$(22) J_{ij} = (J^{ij})^{-1} = \text{cov} \left(\frac{\partial \log \phi}{\partial \lambda_i}, \frac{\partial \log \phi}{\partial \lambda_j} \right)$$

Now we proceed to derive the gamma and beta function inequalities in Section 2. Section

3 gives a matrix variate generalization of Gauss' hypergeometric series formula.

2. Inequalities

Obviously from (19) an unbiased estimate T of $|\Lambda|^\alpha$, $\alpha > 0$, is

$$(23) \quad T = C_\alpha |S|^\alpha, \quad C_\alpha = (\Gamma_p^*(a + \alpha))^{-1} \Gamma_p^*(a),$$

and

$$(24) \quad V(T) = |\Lambda|^{2\alpha} \left(\frac{C_\alpha^2}{C_{2\alpha}} - 1 \right).$$

Further, we have that

$$(25) \quad J_{ij} = \text{cov} \left(\frac{\partial \log \phi}{\partial \lambda_i}, \frac{\partial \log \phi}{\partial \lambda_j} \right) = 0, \quad i \neq j,$$

$$(26) \quad J_{ii} = E \left(- \frac{\partial^2 \log \phi}{\partial \lambda_i^2} \right) = (a_{p-i+1}) \lambda_i^{-2},$$

$$(27) \quad \frac{\partial |\Lambda|^\alpha}{\partial \lambda_i} = \alpha |\Lambda|^\alpha / \lambda_i, \quad i = 1, \dots, p,$$

and hence from (21) we have that

$$(28) \quad [\Gamma_p^{*2}(a + \alpha) / \Gamma_p^*(a) \Gamma_p^*(a + 2\alpha)] \leq \left(1 + \sum_{j=1}^p (a_{p-j+1})^{-1} \alpha^2 \right)^{-1},$$

which is a generalization of (1).

The modified multivariate beta density of a $p \times p$ opdH matrix M is, Olkin (1959),

$$(29) \quad (B_p^*(a, b))^{-1} \prod_{i=1}^p \lambda_i^{-(a_{p-i+1} + b_{p-i+1} - 1)} |M|^{a_{p-2t(p-1)} - 1} \\ |\Lambda - M|^{b_{p-2t(p-1)} - 1} \prod_{j=2}^p |M_j|^{-k_j - 1} |(\Lambda - M)_j|^{-m_j - 1},$$

where $b_p = 2mt$, $b_j = m_{p-j+1} + \dots + m_p$, $m = (m_1, \dots, m_p)$, $b = (b_1, \dots, b_p)$, $b_j > 2t(p-1) + 1$, and

$$(30) \quad B_p^*(a, b) = \Gamma_p^*(a) \Gamma_p^*(b) / \Gamma_p^*(a + b)$$

We use the same procedure as we used to establish (28), and we find that

$$(31) [B_p^*(a+\alpha)/B_p^*(a,b) B_p^*(a+2\alpha, b)] \\ \leq (1 + \alpha^2 \sum_{j=1}^p (a_{p-j+1})^{-1} (a_{p-j+1} + b_{p-j+1} - 1)^{-1})^{-1}$$

Again following Olkin (1959) we write down the second kind of modified multivariate beta density of a $p \times p$ opdH matrix M as

$$(32) (B_p^*(a,b))^{-1} \prod_{i=1}^p \lambda_i^{b_{p-i+1}} |M|^{ap-2t(p-1)-1} |\Lambda + M|^{-(a_p+b_p)} \\ \prod_{j=2}^p |(M(\Lambda + M)^{-1})_j|^{-k_j-1} |((\Lambda + M)^{-1})_j|^{-m_j-1},$$

and find an analog of (31) as

$$(33) [B_p^*(a+\alpha, b-\alpha)/B_p^*(a, b) B_p^*(a+2\alpha, b-2\alpha)] \\ \leq (1 + \alpha^2 \sum_{j=1}^p (a_{p-j+1} b_{p-j+1})^{-1} (a_{p-j+1} + b_{p-j+1} + 1))^{-1}.$$

Now we proceed to derive Gauss' hypergeometric series formula.

3. Hypergeometric Series Formula

Following Constantine (1963) we write the following, slightly modified, definitions

$$(34) \Gamma_p^*(a, \theta) = \pi^{tp(p-1)} \prod_{i=1}^p \Gamma(a_i + \theta_i - 2t(i-1)),$$

$$(35) \Gamma_p^*(a, 0) = \Gamma_p^*(a),$$

$$(36) [a]_\theta = \prod_{j=1}^p \{\Gamma(a_j + \theta_j - 2t(j-1))/\Gamma(a_j - 2t(j-1))\} = \Gamma_p^*(a, \theta)/\Gamma_p^*(a)$$

Now Constantine (1963, p. 1274, equation (15)) derives (34) with $a_1 = a_2 = \dots = a_p = a$ by using the weight function as the Wishart distribution. By using the Bellman-Wishart density $\phi(S, I)$ of (19) as the weight function we see that (34) is true and hence several results given by Constantine (1963) will hold true if in Constantine's (1963) result his $\Gamma_p(a, \theta)$ is replaced by our $\Gamma_p^*(a, \theta)$, where $\theta = (\theta_1, \dots, \theta_p)$ is a partition of θ , $\theta_1 \geq \theta_2 \geq \dots \geq \theta_p \geq 0$, into not more than p parts. Now from Constantine (1963, p. 1284, equation (61)) it follows that, by setting $\Omega = I$,

$$(37) {}_2F_1(\alpha, \beta, \delta, I) = [\Gamma_p(\delta) \Gamma_p(\delta - \alpha - \beta) / \Gamma_p(\delta - \alpha) \Gamma_p(\delta - \beta)],$$

and hence in our notation it must be true that

$$(38) {}_2F_1(\alpha, \beta, \delta, I) = [\Gamma_p^*(\delta) \Gamma_p^*(\delta - \alpha - \beta) / \Gamma_p^*(\delta - \alpha) \Gamma_p^*(\delta - \beta)].$$

Note that when all the components of a in (36) are equal, then (36) reduces to Constantine's (1963, p. 1276, equation (26)) $[a]_\theta$, which he writes as $\Gamma_p(a, \theta) / \Gamma_p(a)$.

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